

Linear SVM Classification on Iris Dataset with Decision Boundary Visualization

Aim:

To implement a **Linear Support Vector Machine (SVM)** for binary classification on the Iris dataset and visualize the **decision boundary and margin**.

Theory

A Linear SVM finds a hyperplane:

$$w \cdot x + b = 0$$

that separates two classes with **maximum margin**.

◆ Margin Concept

$$\text{Margin} = \frac{2}{\|w\|}$$

- Margin = distance between two supporting hyperplanes
- Larger margin → better generalization
- Defined by **support vectors**

Dataset: Iris Dataset

We use the famous Iris dataset.

- 3 classes: Setosa, Versicolor, Virginica
- For simplicity → convert to **binary classification**:
 - Setosa → 1
 - Non-Setosa → 0

We use only **2 features** for visualization:

- Sepal Length
- Sepal Width

Algorithm

1. Load dataset
2. Convert into binary classification
3. Select two features

4. Train Linear SVM
5. Plot decision boundary
6. Plot margin and support vectors

Program

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn import svm, datasets

# Load dataset

iris = datasets.load_iris()

X = iris.data[:, :2] # take first 2 features

y = iris.target

# Convert to binary classification (Setosa vs Non-Setosa)

y = np.where(y == 0, 1, -1)

# Train Linear SVM

model = svm.SVC(kernel='linear', C=1)

model.fit(X, y)

# Plot data points

plt.scatter(X[:, 0], X[:, 1], c=y)

# Plot decision boundary

ax = plt.gca()

xlim = ax.get_xlim()

ylim = ax.get_ylim()
```

```
xx = np.linspace(xlim[0], xlim[1], 100)
yy = np.linspace(ylim[0], ylim[1], 100)
YY, XX = np.meshgrid(yy, xx)
xy = np.vstack([XX.ravel(), YY.ravel()]).T

Z = model.decision_function(xy).reshape(XX.shape)

# Decision boundary
ax.contour(XX, YY, Z, levels=[0])

# Margins
ax.contour(XX, YY, Z, levels=[-1, 1], linestyles=['--', '--'])

# Support vectors
ax.scatter(model.support_vectors_[0],
           model.support_vectors_[1],
           s=100, facecolors='none', edgecolors='k')

plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.title("Linear SVM on Iris Dataset")
plt.show()
```